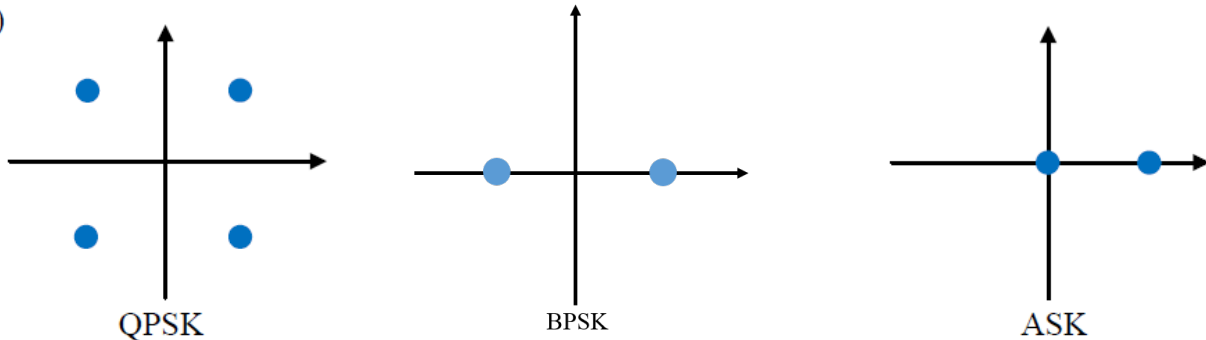


EE558 HW5

Problem 1:

(a)



(b) QPSK: 2 bit/s/Hz

(a) The figure on the left corresponds to NRZ and the figure on the right corresponds to RZ. For the same bit rate, the pulses (i.e., time-domain) in RZ signal are narrower and thus have broader optical spectrum (i.e., frequency-domain).

(b) NRZ signal is more tolerant to dispersion because of the narrow optical spectrum.

(c) Peak (i.e., clock tone) spacing is equal to the bit rate, 10 GHz.

Problem 2:

As a consequence, $BER = 10^{-9}$ and 10^{-3} correspond to $Q=6$ and 3.1.

Problem 3:

$$BER = 10^{-9} \rightarrow Q = 6 = \frac{V_1 - V_0}{\sigma_1 + \sigma_0} = \frac{V_1}{2 \times (50 \text{ nV})} \rightarrow V_1 = 600 \text{ nV} = 0.6 \text{ } \mu\text{V}.$$

Problem 4:

(a) When the cutoff frequency of a LPF is increased, more noise is allowed to pass through the filter and thus signal will become noisier.

(b) When the cutoff frequency is decreased, high-frequency components of the signal are cut off and the signal will suffer from the inter-symbol interference (ISI) and become distorted. However, the signal will have less noise.

Problem 5:

$BER = 10^{-9} \rightarrow Q = 6$. For a given BER, $N_p = 71.41/\eta$ (ASK-Heterodyne), $35.95/\eta$ (ASK-Homodyne), $17.97/\eta$ (PSK-Heterodyne), and $8.48/\eta$ (PSK-Homodyne). If the quantum efficiency $\eta \approx 1$, the PSK (Homodyne) requires around 9 photons/bit, showing the most sensitivity.